

MedVLA: A Hierarchical Vision-Language-Action Framework for Closed-Loop Precision Medical Robot Manipulation

Junjie Xie, Angen Ye, Yujia Song, Chuxuan He, and Dapeng Zhang*

Abstract—Precision medical robotics demands adaptive decision-making under strict safety, interpretability, and execution constraints. Although recent Vision-Language-Action (VLA) models show strong multimodal reasoning ability, their continuous action generation paradigm is not well suited for precision medical tasks, where reliable closed-loop operation may also depend on non-action system function calls. To address this gap, we propose MedVLA, a hierarchical framework that couples high-level multimodal reasoning with low-level function-constrained execution. We further introduce a scalable multi-agent pipeline to generate skill-oriented chain-of-thought(CoT) data for structured training. Built on different multimodal large-model backbones, MedVLA consistently improves performance after fine-tuning, demonstrating the effectiveness of the proposed framework across model variants. Under identical initial conditions, we perform 100 closed-loop flexible electrode implantation trials. The results show that MedVLA achieves a 95.0% task success rate, substantially outperforming representative VLA baselines, including OpenVLA (8%) and π_0 (15%), in accuracy, stability, and safety. These results indicate that structured reasoning with constrained function-level execution is a practical route toward deployable precision medical robotics.

I. INTRODUCTION

Precision medical robotics has become an important research direction at the intersection of medical engineering, robotics, and intelligent systems [1]. Its goal is not merely to complete predefined operations, but to maintain stable control of instruments and tissues in medical environments characterized by high risk, strong constraints, and extremely low fault tolerance. Unlike general manipulation in open environments, precision medical tasks are typically performed in narrow surgical fields and require continuous perception, real-time judgment, and dynamic correction within millimeter-scale or smaller workspaces [2]–[5]. In scenarios such as minimally invasive intervention, neural modulation, and electrode implantation, the robot must not only localize key targets from visual observations, but also continuously update subsequent actions by jointly considering instrument state, task phase, and operation goals, thereby enabling safe and accurate closed-loop execution under continuous feedback [4]–[6]. Such settings therefore require integrated

high-precision decision-making, constrained execution, and closed-loop adjustment under complex operational constraints [5], [6].

Medical robotics has made sustained progress in mechanism design, visual navigation, motion control, and human-robot collaboration [3], [4], [7], [8]. However, most existing systems still remain within the “human-led, robot-assisted” paradigm, where intelligence is reflected in local functional enhancement rather than autonomous decision-making throughout the task process [9]–[12]. Meanwhile, Vision-Language-Action (VLA) models such as OpenVLA and π_0 have unified visual perception, language understanding, and action generation within a single framework, showing strong cross-task generalization in open-world manipulation [13], [14]. Yet a substantial gap remains between such general-purpose VLA paradigms and precision medical requirements. Schmidgall *et al.* pointed out that robot-assisted surgery faces distinctive challenges including limited open surgical data, soft-tissue modeling difficulty, and stricter safety validation needs [15]. Lee *et al.* confirmed through a review of FDA-cleared surgical robots that most deployed systems operate at low autonomy levels [9]. Zargarzadeh *et al.* further showed that in surgical settings, higher-level reasoning must be explicitly coordinated with lower-level execution rather than left to unconstrained end-to-end action generation [16]. These findings indicate that directly applying general-purpose VLA models to precision medical tasks remains insufficient in execution accuracy, safety constraints, and behavioral determinism. Beyond the execution-level mismatch, data acquisition has become another major bottleneck. Previous studies have relied on small-scale experimental data or task-specific samples collected on dedicated platforms, which remain insufficient for systems that must learn high-level reasoning and closed-loop feedback mechanisms [6], [15], [17], [18]. Data collection for precision medical tasks is inherently costly due to specialized platforms, dense annotation, and expert participation [8], [12], [19]. Moreover, recent work shows that raw visual input with direct supervision cannot capture the full decision chain of closed-loop medical operation; richer representations such as hierarchical phase-task-action annotations [20], explicit phase recognition [21], and structured surgical workflow modeling [17], [22], [23] are necessary. What is therefore needed is a scalable data construction mechanism that can generate structured task data efficiently while remaining extensible to new precision medical scenarios [6], [18].

To address these issues, this paper proposes **MedVLA**, a hierarchical vision-language-action framework for closed-

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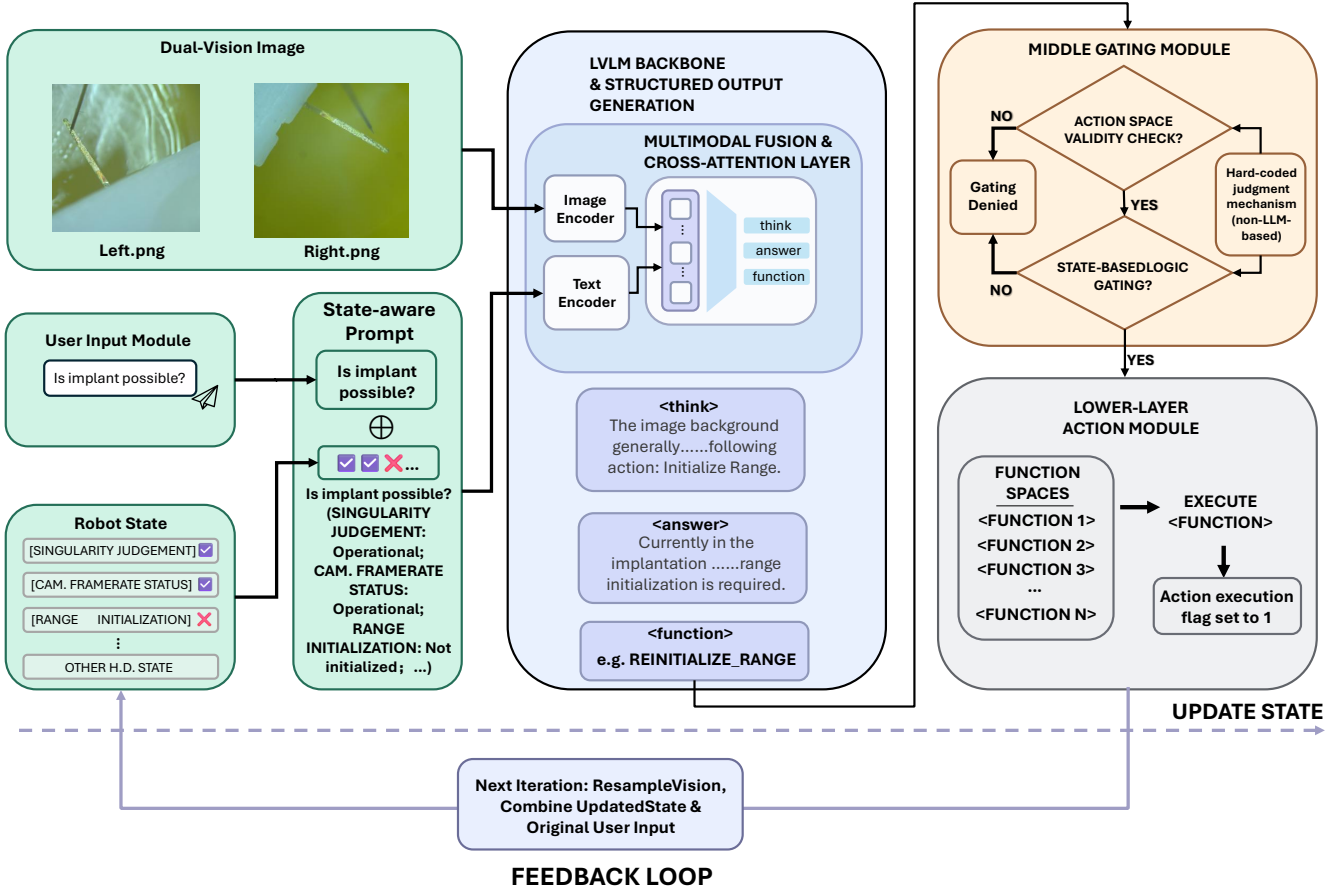


Fig. 1. Overview of the proposed hierarchical closed-loop framework for precision medical robot operation. Dual-view images, user instruction, and robot state are combined into a state-aware prompt and processed by the LVLN to generate structured outputs (think, answer, and function). The predicted function is then verified by the middle gating module through action-space validity checking and state-based logic gating before being executed by the lower-layer action module. Updated visual observations and robot states are fed back into the next iteration to form a closed-loop process.

loop precision medical robot manipulation. While hierarchical control and skill-based planning have been explored in general robotics [13], [14], MedVLA differs from prior work in three key respects. First, unlike high-level planner / low-level controller frameworks that decompose tasks into language-level subgoals and delegate execution to learned continuous policies, MedVLA outputs structured function calls verified by an explicit state-based gating module before reaching the executor, making each decision auditable and constrained rather than open-ended. Second, the gating mechanism enforces domain-specific preconditions tied to the physical task state—not merely action-space membership—so that unsafe or premature operations are blocked before execution. Third, the framework integrates chain-of-thought reasoning into the decision triplet, coupling interpretable reasoning traces with function prediction to support both post-hoc analysis and improved intermediate judgment accuracy. Together, these design choices address the safety, determinism, and verifiability requirements that distinguish precision medical manipulation from open-world robotic tasks.

Specifically, the upper module is centered on a fine-tuned

large model that integrates visual observations, system states, and task goals to produce structured decision outputs. The lower module invokes constrained skill interfaces according to these decisions, executes them in a single-step manner, updates the environment state after each operation, and feeds the updated state back to the upper module, thereby forming a progressively advancing closed-loop process. Through this design, MedVLA preserves the high-level semantic understanding and task reasoning ability of large models [13], [14] while strictly restricting actual execution to a constrained and verifiable skill space, making it better suited to the safety, determinism, and execution-precision requirements of precision medical tasks [1], [4]. In addition, to address the high cost and limited scalability of real-world medical data collection, we design a skill-oriented multi-agent data generation mechanism for MedVLA. Multiple agents collaboratively construct instruction, state, decision, and skill-invocation data throughout the task process, while a cosine-similarity-based deduplication strategy removes highly repetitive samples. This process yields a structured dataset with lower redundancy, stronger generalization ability, and clearer organization for MedVLA training and deployment.

We conduct systematic validation of the proposed method on a real implantable electrode operation platform. The primary contributions of this work are summarized as follows:

1) We propose a hierarchical architecture tailored for precision medical tasks, which achieves safe closed-loop control over a constrained skill space by decoupling high-level judgment from low-level execution.

2) We design a skill-oriented multi-agent data generation mechanism and construct a structured dataset of approximately 30,000 samples with low redundancy and strong generalization, providing an effective data foundation for closed-loop decision learning.

3) We perform system-level validation on a real platform and show that the proposed method achieves a 95.0% success rate in complete closed-loop tasks, outperforming OpenVLA and π_0 by 87.0 and 80.0 percentage points, respectively, thereby demonstrating its effectiveness in balancing precision execution and intelligent decision-making.

II. METHODOLOGY

A. Task Formulation and Problem Modeling

Precision medical robotic tasks are structured decision-making problems under multimodal and state constraints, rather than direct regressions from visual observations to continuous actions. Compared with general manipulation, they involve stringent precision requirements and explicit execution stages. At each step, the system decision depends on visual observations, execution progress, completed actions, and task objectives. We therefore formulate the task as an iterative process centered on state-constrained function calls.

At time t , the system receives binocular image observations $I_t = \{I_t^{(1)}, I_t^{(2)}\}$, user task input u_t , and internal robot state s_t . Here, I_t captures the surgical scene and local geometric relations, u_t specifies the task intent, and s_t records the execution stage and internal decision variables. Based on these inputs, the system predicts a structured decision triplet

$$y_t = \{\langle \text{think} \rangle_t, \langle \text{answer} \rangle_t, \langle \text{function} \rangle_t\}, \quad (1)$$

where $\langle \text{think} \rangle_t$ denotes the reasoning process, $\langle \text{answer} \rangle_t$ the intermediate judgment, and $\langle \text{function} \rangle_t$ the next function call. The core mapping learned in this work is

$$(I_t, u_t, s_t) \rightarrow y_t, \quad (2)$$

rather than the conventional mapping $(I_t, u_t) \rightarrow a_t$.

Let $\mathcal{F}$ denote the discrete executable function space, and let $\mathcal{F}(s_t) \subseteq \mathcal{F}$ denote the subset allowed under the current state. The predicted function must satisfy

$$\langle \text{function} \rangle_t \in \mathcal{F}(s_t). \quad (3)$$

The overall task is thus formulated as the closed-loop process

$$(I_t, u_t, s_t) \rightarrow y_t \rightarrow \langle \text{function} \rangle_t \rightarrow (s_{t+1}, I_{t+1}), \quad (4)$$

which repeats until termination.

B. Hierarchical Closed-Loop Control Framework

To ensure stable execution under stringent constraints, we design a hierarchical closed-loop framework that organizes perception, reasoning, verification, and execution into a unified workflow. As shown in Fig. 1, the framework consists of multimodal perception and state inputs, a high-level decision module, a middle gating module, a low-level action module, and a feedback update loop.

At each step, the robot state is explicitly combined with the user instruction to form a state-aware prompt. This allows the model to reason not only about the visible scene, but also about what is currently executable under the system state. The high-level decision module uses a fine-tuned Large Vision-Language Model (LVLM) to map the image observations and state-aware prompt to the structured triplet y_t defined in Eq. (1), making the decision process explicit and interpretable.

To prevent invalid or unsafe execution, the predicted $\langle \text{function} \rangle_t$ is checked by the middle gating module, which performs two sequential checks: (i) *action-space validity*, ensuring that the predicted function belongs to the predefined executable set $\mathcal{F}$; and (ii) *state-based logic gating*, verifying that the function is consistent with the current robot state s_t and satisfies domain-specific preconditions. In the implantation task, these preconditions include geometric alignment status between the needle and electrode hole, vision-system readiness (exposure and focus within operational bounds), and workspace initialization state. Only functions that pass both checks are forwarded to the lower layer for execution.

When the gate rejects a predicted function, the system immediately triggers `E_STOP`, initiates a hardware self-check routine, and activates an audible alarm. Given the safety-critical nature of precision surgical tasks, the system then transfers control to the human operator rather than attempting autonomous recovery to an uncertain state. This conservative design reflects the principle that, in high-stakes medical settings, a safe halt with human handover is preferable to autonomous re-planning whose safety cannot be fully guaranteed.

After execution, the system updates the robot state and visual observations, yielding (s_{t+1}, I_{t+1}) for the next iteration. This forms an iterative “Perceive–Reason–Gate–Execute–Update” loop, summarized in Algorithm 1.

C. Structured Data Generation via Multi-Agent Collaboration

To align the training data with the proposed framework, we construct a structured data generation pipeline based on multi-agent collaboration, redundancy reduction, and constraint verification, as illustrated in Fig. 2.

We use prior task documents, robot capability documents, action rules, and initial system states as knowledge sources for data generation. In this pipeline, we define two LLM-based roles: the *Human Agent*, which simulates a user by generating diverse task prompts from the source materials, and the *Robot Agent*, which produces structured responses given the generated prompts and system states. The term

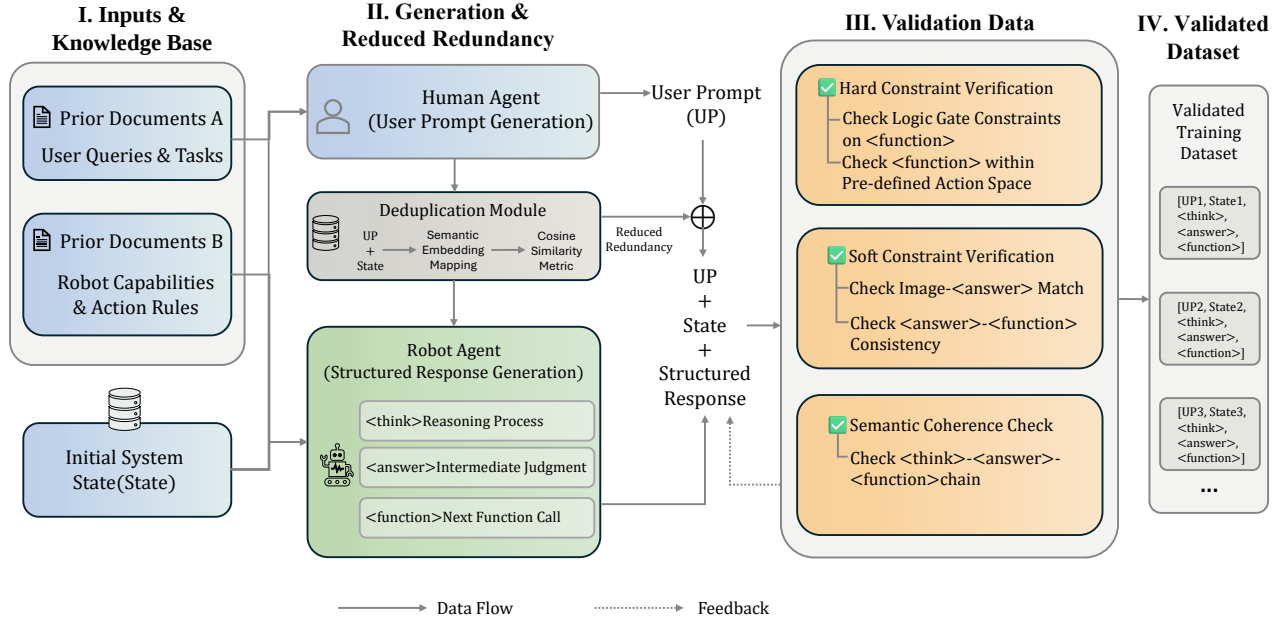


Fig. 2. Structured data generation pipeline based on multi-agent collaboration. Prior documents and initial system states are used to generate user prompts. The combined prompt-state inputs are embedded using the hidden representations of the LVLM backbone, and redundant samples are removed by cosine similarity. The retained samples are then used to generate structured responses (*think*, *answer*, and *function*) and further filtered by hard constraints, soft constraints, and semantic coherence checks to form the validated training dataset.

Algorithm 1 Hierarchical Closed-Loop Control

Require: Images I_0 , user input u , initial state s_0 , LVLM $\mathcal{M}$, function space $\mathcal{F}$

- 1: $t \leftarrow 0$
- 2: **while** task not terminated **do**
- 3: $y_t \leftarrow \mathcal{M}(I_t, u, s_t)$ {Reason: predict triplet}
- 4: $f_t \leftarrow \langle \text{function} \rangle_t$
- 5: **if** $f_t \notin \mathcal{F}$ **or** $f_t \notin \mathcal{F}(s_t)$ **then**
- 6: Execute `E_STOP`; self-check; alarm; hand over {Gate: reject}
- 7: **end if**
- 8: Execute f_t ; update (s_{t+1}, I_{t+1}) {Execute & Update}
- 9: $t \leftarrow t + 1$
- 10: **end while**

skill-oriented refers to the fact that each training sample is organized around a single executable function call (i.e., a predefined robot skill) rather than free-form text or raw action trajectories. The Human Agent first generates user prompts to simulate realistic task requests. To reduce redundancy, each prompt-state pair is embedded using the internal hidden representations of Qwen2.5-VL-3B-Instruct, and highly similar samples are removed by cosine similarity thresholding.

The retained prompt-state pairs are then fed to the Robot Agent, which generates a structured response following the same triplet format as Eq. (1). The generated response is paired with the corresponding prompt and state for verification. The Verification Agent then performs three levels of checking: hard constraint verification, which tests whether the predicted function satisfies logic-gate constraints and remains within the predefined action space; soft constraint verification, which checks consistency between image obser-

vations and $\langle \text{answer} \rangle$, as well as alignment between $\langle \text{answer} \rangle$ and $\langle \text{function} \rangle$; and semantic coherence checking over the full $\langle \text{think} \rangle$ - $\langle \text{answer} \rangle$ - $\langle \text{function} \rangle$ chain. Only samples that pass all checks are retained to form the final validated training dataset.

This pipeline improves data quality by jointly reducing redundancy and enforcing constraint consistency, producing training data better aligned with the state-aware, function-constrained, and closed-loop characteristics of precision medical robotic tasks.

III. EXPERIMENTS

A. Experimental Platform, Setup, and Evaluation Protocol

We evaluate the proposed framework on a real-world flexible electrode implantation platform for precision neurosurgical manipulation, where a tungsten insertion needle ($\sim 10 \mu\text{m}$ tip) must pass through a $30 \times 50 \mu\text{m}$ electrode hole (Fig. 3). The micrometer-scale geometric constraints make this task representative of precision surgical robotic systems. We conduct both offline function prediction and real-robot closed-loop evaluation.

The decision space consists of seven discrete functions, grouped into image regulation, state management, safety control, and action execution, as listed in Table I. Among them, `ALIGN_NEEDLE` and `INSERT` are the two execution-critical functions directly associated with implantation, while the others provide visual adjustment, system recovery, and safe interruption during closed-loop operation.

We organize the evaluation into offline function prediction and real-robot closed-loop execution. For dataset construction, we collect approximately 3,000 real-world samples

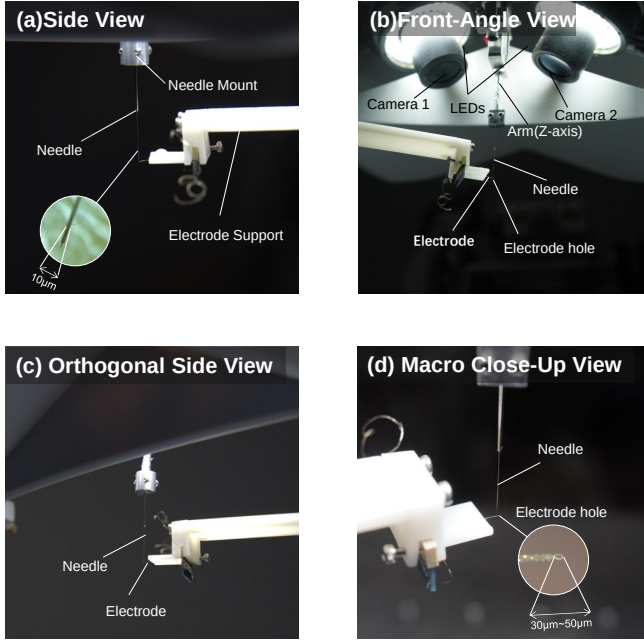


Fig. 3. Experimental platform for flexible electrode implantation. (a) Side view of the needle mount, insertion needle, and electrode support, with a zoom-in of the $\sim 10 \mu\text{m}$ needle tip. (b) Front-angle view of the overall setup, including the binocular cameras, LEDs, robotic arm, and flexible electrode. (c) Orthogonal side view of the needle–electrode configuration. (d) Macro close-up of the insertion region, showing alignment between the needle and the approximately $30 \times 50 \mu\text{m}$ electrode hole.

(synchronized binocular images, function labels, and trajectories at 5 Hz) and expand to $\sim 30,000$ samples through controlled image perturbations. Combined with the data generation pipeline in Section II, this provides a unified training foundation for all methods. To verify backbone generality, we fine-tune InternVL2-2B, Qwen2.5-VL-3B, and Llama-3.2-11B-Vision-Instruct. For comparison, we evaluate OpenVLA-7B and π_0 -3.3B in real-robot closed-loop execution. Qwen2.5-VL-3B is selected for deployment due to its favorable trade-off between accuracy and efficiency. We report **Exact Match (EM)** and **Macro-F1** for offline evaluation, and closed-loop insertion success rate for real-robot evaluation.

B. Offline Function Prediction Results

Table II reports offline function prediction results under zero-shot evaluation, cross-backbone fine-tuning (i.e., applying the same structured training procedure to LVLM backbones other than the primary Qwen2.5-VL), and CoT ablation. Overall, the full Qwen2.5-VL-3B variant reaches 95.00% accuracy and 0.9604 Macro-F1, substantially outperforming the zero-shot baselines, which supports the effectiveness of structured function-level supervision for precision surgical decision making.

Comparing *Ours w/o CoT* and *Ours Full* shows that the effect of CoT supervision is class-dependent. The largest gains appear on `ADJUST_FOCUS` (65.84% $\rightarrow$ 96.71%) and `ALIGN_NEEDLE` (48.67% $\rightarrow$ 72.57%), indicating that CoT is particularly helpful for functions requiring finer judgment

TABLE I

DISCRETE FUNCTION SPACE USED IN THE PROPOSED FRAMEWORK. SYMBOLS INDICATE FUNCTION CATEGORIES: $\blacktriangle$ IMAGE REGULATION, $\star$ STATE MANAGEMENT, $\blacksquare$ SAFETY CONTROL, AND $\bullet$ ACTION EXECUTION.

Function	Description
$\blacktriangle$ <code>ADJUST_EXPOSURE</code>	Adjust image exposure to improve brightness and visibility.
$\blacktriangle$ <code>ADJUST_FOCUS</code>	Adjust focus to improve scene sharpness.
$\star$ <code>REINITIALIZE_RANGE</code>	Reset the robot working range when the current state is unsuitable.
$\star$ <code>RESTART_CAMERA</code>	Restart the vision system after camera failure or abnormal observation.
$\blacksquare$ <code>E_STOP</code>	Immediately stop execution for safety protection.
$\bullet$ <code>ALIGN_NEEDLE</code>	Refine the alignment between the insertion needle and the target.
$\bullet$ <code>INSERT</code>	Execute the final insertion toward the target position.

of visual quality and intermediate surgical states. By contrast, `INSERT` decreases from 93.33% to 81.67%. Error analysis on the 60 `INSERT` test samples reveals that the additional false negatives introduced by CoT are predominantly misclassified as `ALIGN_NEEDLE`, meaning the model favors further alignment refinement rather than committing to insertion when the reasoning chain expresses residual uncertainty about geometric readiness. This conservative shift is not necessarily harmful: as shown in Section III-C, permitting additional alignment rounds in closed-loop execution leads to a higher overall insertion success rate (95% vs. 73% at 3 rounds), because the real-world cost of a redundant `ALIGN_NEEDLE` call is far lower than the cost of a premature `INSERT`. Despite this per-class reduction, CoT improves reasoning over ambiguous intermediate states and raises overall accuracy from 85.30% to 95.00%.

Cross-backbone results further show that the framework generalizes well across different LVLM architectures. Under the same structured function-calling setting, both InternVL2-2B and Llama-3.2-11B achieve competitive results, indicating that the framework does not depend on a specific backbone design. The Llama-based variant attains the best overall accuracy, likely due to its larger parameter scale. However, Qwen2.5-VL-3B remains highly competitive with a much smaller model size and offers a better practical trade-off for deployment, and is therefore selected for real-robot experiments.

As shown in Table II, the remaining errors of the full model are mainly concentrated in three functions: `ADJUST_FOCUS`, `ALIGN_NEEDLE`, and `INSERT`. By contrast, `ADJUST_EXPOSURE`, `REINITIALIZE_RANGE`, `RESTART_CAMERA`, and `E_STOP` are predicted with near-perfect or perfect accuracy, indicating that the major difficulty lies in visually ambiguous intermediate judgments and

TABLE II
OFFLINE FUNCTION PREDICTION PERFORMANCE OF DIFFERENT BACKBONES AND ABLATION VARIANTS

Overall Performance on the Action-Balanced Test Set						
Type	Model	Scale	FT	CoT	EM (%)	Macro-F1
Zero-shot	Qwen2.5-VL	3B	No	No	29.60	0.1594
Zero-shot	Qwen3-VL	30B-A3B	No	No	65.40	0.7474
Cross-backbone	InternVL2	2B	Yes	Yes	94.00	0.9400
Cross-backbone	Llama-3.2-11B	11B	Yes	Yes	95.70	0.9570
Ours	Qwen2.5-VL	3B	Yes	No	85.30	0.8693
Ours	Qwen2.5-VL	3B	Yes	Yes	95.00	0.9604

Per-Function Accuracy (%)							
Function (GT)	Support	Qwen2.5-3B	Qwen3-30B	InternVL2-2B	Llama-3.2-11B	Ours w/o CoT	Ours Full
ADJUST_EXPOSURE	243	100.00	84.36	99.18	100.00	99.18	100.00
ADJUST_FOCUS	243	0.00	20.99	95.88	97.12	65.84	96.71
REINITIALIZE_RANGE	141	0.00	100.00	100.00	100.00	100.00	100.00
ALIGN_NEEDLE	113	0.00	39.82	70.80	76.11	48.67	72.57
RESTART_CAMERA	100	1.00	96.00	100.00	100.00	100.00	100.00
E_STOP	100	52.00	100.00	100.00	100.00	100.00	100.00
INSERT	60	0.00	26.67	75.00	85.00	93.33	81.67
OVERALL	1000	29.60	65.40	94.00	95.70	85.30	95.00

precision-critical execution decisions rather than recovery or safety-related functions.

C. Closed-Loop Comparison in Real Execution

We evaluate all methods under a unified 100-trial closed-loop protocol. In each trial, the insertion needle is randomly initialized, and successful insertion is the final criterion. For each initialization, the VLA baseline runs first with all motor trajectories recorded; the robot is then restored to the same start state for our method, ensuring a consistent comparison. Under this protocol, OpenVLA achieves **8** successful insertions out of 100 trials, corresponding to **8.00%**, while π_0 achieves **15** successful insertions, corresponding to **15.00%**. In contrast, the proposed hierarchical framework performs substantially better. With the maximum number of decision rounds limited to **3**, our method achieves **73.00%**; increasing the limit to **6** further improves the success rate to **95.00%**. This result is consistent with the offline observation that the model is more conservative before committing to the final INSERT action. Although such conservativeness may slightly reduce offline INSERT accuracy, it appears to avoid premature or risky insertion attempts in real execution, thereby improving overall success in high-precision electrode implantation.

We chose OpenVLA and π_0 as baselines because they represent the two dominant VLA paradigms—autoregressive action tokenization and flow-matching continuous control—and comparing against them isolates the benefit of hierarchical function-level execution over end-to-end continuous action generation. Conventional visual-servoing or PID/MPC controllers could serve as domain-specific baselines but are limited to low-level subtasks and lack the high-level multimodal reasoning that the proposed framework targets; a fair comparison would require designing equivalent task-level decision logic around them, which we leave for future work.

Inference latency. In our deployment configuration (Qwen2.5-VL-3B on a single GPU), the average end-to-end latency per decision cycle is approximately 10 seconds, comprising ~ 8 s for LVLM inference (including image encoding and structured output generation), ~ 2 s for state update and gating verification, and negligible time for low-level motor command execution. We note that the 5 Hz in Section III-A is the offline trajectory-sampling rate used to build dense training data, and is distinct from the deployment decision rate: at runtime the system issues one structured decision per ~ 10 s cycle (~ 0.1 Hz), and the executed function moves the needle by a single constrained step before the next observation is taken. Since the implantation task operates quasi-statically with no moving tissue, this decision rate is acceptable for the current application. For time-critical scenarios, latency could be reduced via distillation, quantization, or speculative decoding.

Gating module analysis. To assess the role of the middle gating module, we conducted over 500 inference trials on the test set under varied prompt formulations. Across all trials, the gate rejected the predicted function only twice, both due to the LVLM generating outputs outside the predefined seven-function action space. In each case, the system immediately triggered E_STOP, executed a hardware self-check, and activated an audible alarm before handing control to the human operator. The low rejection rate indicates that the fine-tuned LVLM has learned to produce well-formed function calls with high reliability. Nevertheless, the gating module remains essential as a hard safety boundary: because removing it would allow arbitrary model outputs to be directly executed on micrometer-scale surgical hardware, a full ablation (i.e., disabling the gate during physical execution) was not conducted to avoid potential equipment damage. The gating module thus serves as a necessary safeguard whose value lies not in frequent activation but in guaranteeing safe

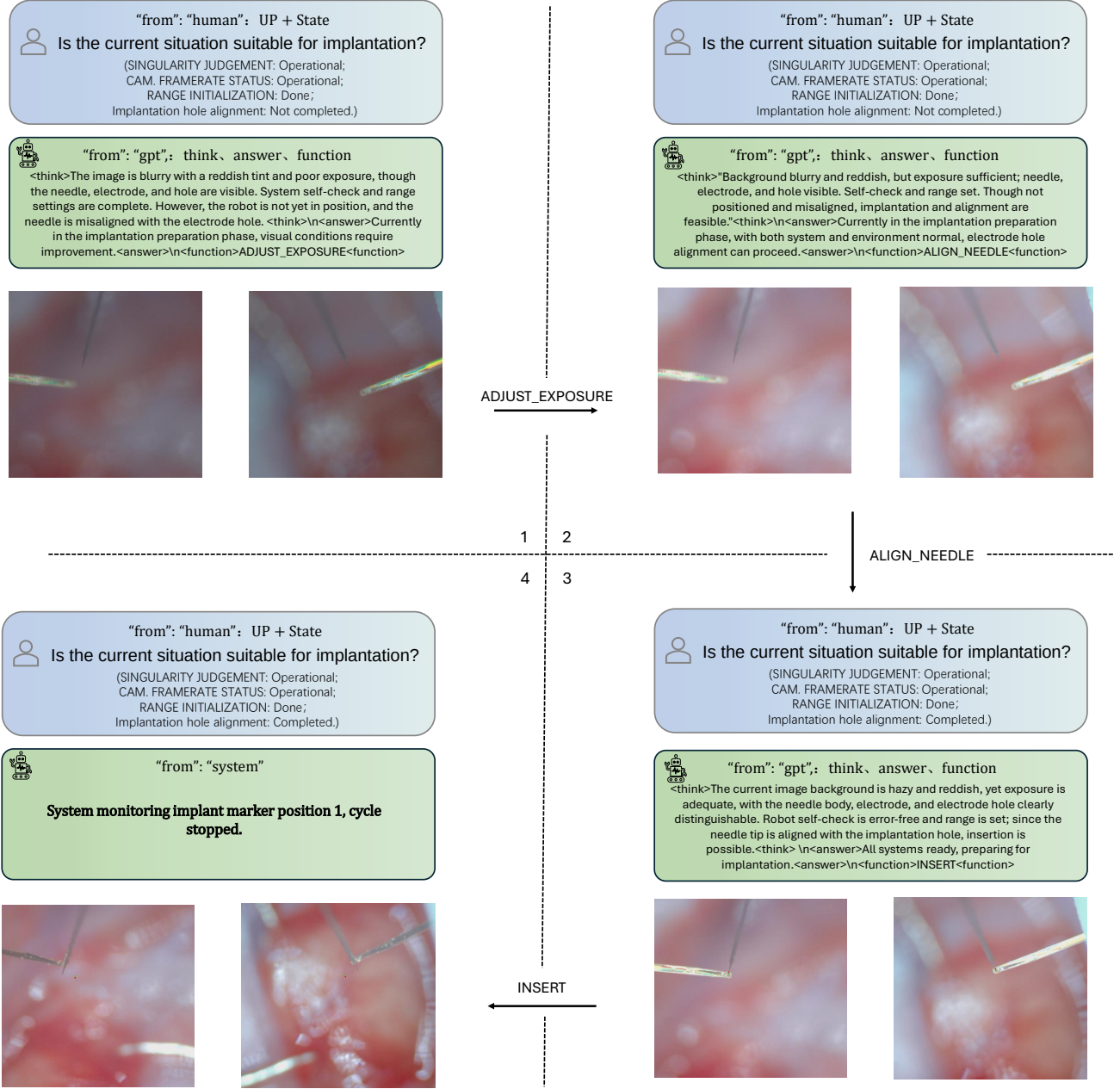


Fig. 4. Real-world closed-loop deployment example of the proposed hierarchical framework using the fine-tuned Qwen2.5-VL-3B-Instruct model. Starting from binocular microscopic observations and task prompts, the system sequentially performs **ADJUST_EXPOSURE**, **ALIGN_NEEDLE**, and **INSERT**, ultimately completing electrode-hole insertion through structured function-level decision making.

termination under rare but critical failure modes.

D. Real-World Closed-Loop Case Study

Fig. 4 shows a representative deployment case using the fine-tuned Qwen2.5-VL-3B-Instruct model. Starting from binocular microscopic observations, the system sequentially issues **ADJUST_EXPOSURE**, **ALIGN_NEEDLE**, and **INSERT**, completing electrode-hole insertion through structured closed-loop decision making.

IV. CONCLUSION

This paper presents a hierarchical vision-language-action framework for precision medical robotics and validates it on flexible electrode implantation under micrometer-level constraints. By decoupling high-level multimodal decision-making from constrained low-level execution, the proposed method achieves a 95% closed-loop success rate, addressing a key limitation of continuous-action VLA paradigms in safety-critical settings. The multi-agent data pipeline further provides scalability: new task actions can be supported by

extending the pipeline and fine-tuning.

Limitations. The current evaluation is conducted on a single precision medical task with training data generated by the proposed pipeline, which may favor the framework by construction. Validation on independent datasets, alternative surgical tasks, and data collected by external operators would strengthen the generalization claim. Additionally, the gating mechanism currently relies on task-specific state-based rules; incorporating richer medical-domain constraints (e.g., anatomical boundaries and tissue-interaction models) remains an important direction for broader clinical applicability.

REFERENCES

- [1] D. K. Iakovidis, P. Vartholomeos, A. L. Gall, M. Cianchetti, O. Ozioko, R. Dahiya, E. Mazomenos, F. Vasconcelos, D. Stoyanov, J. Philpott *et al.*, “Medical & healthcare robotics: a roadmap for enhanced precision, safety, and efficacy,” *Measurement Science and Technology*, vol. 36, no. 10, p. 103001, 2025.
- [2] D. Wu, R. Zhang, A. Pore, D. Dall’Alba, X. T. Ha, Z. Li, Y. Zhang, F. Herrera, M. Ourak, W. Kowalczyk *et al.*, “A review on machine learning in flexible surgical and interventional robots: Where we are and where we are going,” *Biomedical Signal Processing and Control*, vol. 93, p. 106179, 2024.
- [3] Y. Liu, X. Wu, Y. Sang, C. Zhao, Y. Wang, B. Shi, and Y. Fan, “Evolution of surgical robot systems enhanced by artificial intelligence: a review,” *Advanced Intelligent Systems*, vol. 6, no. 5, p. 2300268, 2024.
- [4] G. Dagnino and D. Kundrat, “Robot-assistive minimally invasive surgery: trends and future directions,” *International Journal of Intelligent Robotics and Applications*, vol. 8, no. 4, pp. 812–826, 2024.
- [5] S. Huang, C. Lou, Y. Zhou, Z. He, X. Jin, Y. Feng, A. Gao, and G.-Z. Yang, “Mri-guided robot intervention—current state-of-the-art and new challenges,” *Med-X*, vol. 1, no. 1, p. 4, 2023.
- [6] L. Maier-Hein, S. S. Vedula, S. Speidel, N. Navab, R. Kikinis, A. Park, M. Eisenmann, H. Feussner, G. Forestier, S. Giannarou *et al.*, “Surgical data science for next-generation interventions,” *Nature Biomedical Engineering*, vol. 1, no. 9, pp. 691–696, 2017.
- [7] G. P. Moustris, S. C. Hiridis, K. M. Deliparaschos, and K. M. Konstantinidis, “Evolution of autonomous and semi-autonomous robotic surgical systems: a review of the literature,” *The international journal of medical robotics and computer assisted surgery*, vol. 7, no. 4, pp. 375–392, 2011.
- [8] I. Rivas-Blanco, C. J. Perez-Del-Pulgar, I. Garcia-Morales, and V. F. Munoz, “A review on deep learning in minimally invasive surgery,” *IEEE Access*, vol. 9, pp. 48 658–48 678, 2021.
- [9] A. Lee, T. S. Baker, J. B. Bederson, and B. I. Rapoport, “Levels of autonomy in fda-cleared surgical robots: a systematic review,” *NPJ Digital Medicine*, vol. 7, no. 1, p. 103, 2024.
- [10] C. Varghese, E. M. Harrison, G. O’Grady, and E. J. Topol, “Artificial intelligence in surgery,” *Nature medicine*, vol. 30, no. 5, pp. 1257–1268, 2024.
- [11] B. Vasey, K. A. Lippert, D. Z. Khan, M. Ibrahim, C. H. Koh, H. L. Horsfall, K. S. Lee, S. Williams, H. J. Marcus, and P. McCulloch, “Intraoperative applications of artificial intelligence in robotic surgery: a scoping review of current development stages and levels of autonomy,” *Annals of Surgery*, vol. 278, no. 6, pp. 896–903, 2023.
- [12] N. Kenig, J. Monton Echeverria, and A. Muntaner Vives, “Artificial intelligence in surgery: a systematic review of use and validation,” *Journal of clinical medicine*, vol. 13, no. 23, p. 7108, 2024.
- [13] M. J. Kim, K. Pertsch, S. Karamcheti, T. Xiao, A. Balakrishna, S. Nair, R. Rafailov, E. Foster, G. Lam, P. Sanketi *et al.*, “Openvla: An open-source vision-language-action model,” *arXiv preprint arXiv:2406.09246*, 2024.
- [14] K. Black, N. Brown, D. Driess, A. Esmail, M. Equi, C. Finn, N. Fusai, L. Groom, K. Hausman, B. Ichter *et al.*, “ π_0 : A vision-language-action flow model for general robot control,” *arXiv preprint arXiv:2410.24164*, 2024.
- [15] S. Schmidgall, J. W. Kim, A. Kuntz, A. E. Ghazi, and A. Krieger, “General-purpose foundation models for increased autonomy in robot-assisted surgery,” *Nature Machine Intelligence*, vol. 6, no. 11, pp. 1275–1283, 2024.
- [16] S. Zargarzadeh, M. Mirzaei, Y. Ou, and M. Tavakoli, “From decision to action in surgical autonomy: Multi-modal large language models for robot-assisted blood suction,” *IEEE Robotics and Automation Letters*, vol. 10, no. 3, pp. 2598–2605, 2025.
- [17] Y. Li, Z. Zhao, R. Li, and F. Li, “Deep learning for surgical workflow analysis: a survey of progresses, limitations, and trends,” *Artificial Intelligence Review*, vol. 57, no. 11, p. 291, 2024.
- [18] L. Maier-Hein, M. Eisenmann, D. Sarikaya, K. März, T. Collins, A. Malpani, J. Fallert, H. Feussner, S. Giannarou, P. Mascagni *et al.*, “Surgical data science—from concepts toward clinical translation,” *Medical image analysis*, vol. 76, p. 102306, 2022.
- [19] K. Sone, S. Tanimoto, Y. Toyohara, A. Taguchi, Y. Miyamoto, M. Mori, T. Iriyama, O. Wada-Hiraike, and Y. Osuga, “Evolution of a surgical system using deep learning in minimally invasive surgery,” *Biomedical reports*, vol. 19, no. 1, p. 45, 2023.
- [20] M. Bilal, M. Alam, D. Bapu, S. Korsgen, N. Lal, S. Bach, A. M. Hajiyavand, M. Ali, K. Soomro, I. Qasim *et al.*, “Temset-24k: Densely annotated dataset for indexing multipart endoscopic videos using surgical timeline segmentation,” *Scientific Data*, vol. 12, no. 1, p. 1424, 2025.
- [21] F. R. Kolbinger, S. Bodenstedt, M. Carstens, S. Leger, S. Krell, F. M. Rinner, T. P. Nielsen, J. Kirchberg, J. Fritzmann, J. Weitz *et al.*, “Artificial intelligence for context-aware surgical guidance in complex robot-assisted oncological procedures: An exploratory feasibility study,” *European Journal of Surgical Oncology*, vol. 50, no. 12, p. 106996, 2024.
- [22] M. Wagner, B.-P. Müller-Stich, A. Kisilenko, D. Tran, P. Heger, L. Mündermann, D. M. Lubotsky, B. Müller, T. Davitashvili, M. Capek *et al.*, “Comparative validation of machine learning algorithms for surgical workflow and skill analysis with the heichole benchmark,” *Medical image analysis*, vol. 86, p. 102770, 2023.
- [23] P. Li, X. Shu, C.-M. Feng, Y. Feng, W. Zuo, and J. Tang, “Surgical video workflow analysis via visual-language learning,” *npj Health Systems*, vol. 2, no. 1, p. 5, 2025.